THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 1

Chapter 10: **Tissues in Action** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** similar cells	Same function.
2	**(d)** epithelial tissue	Covering.
3	**(a)** hinge joint	One plane.
4	**(b)** meristematic	Cell division.
5	**(a)** nervous	Neurons.
6	**(d)** ball and socket joint	All directions.
7	**(d)** xylem	Upward.
8	**(c)** muscular	Contraction.
9	**(d)** fixed joints	No movement.
10	**(c)** phloem	Sieve tubes.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Group of similar cells performing a function (2).	2
12	Conducting tissue carrying food (2).	2
13	Tissue connecting bone to bone (2).	2
14	Meristematic (1); Permanent (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Apical (1); Lateral (1); Intercalary (1).	1+1+1
16	Transport (1); Defence (1); Clotting (1).	1+1+1
17	Receive messages (1); Carry messages (1); Coordinate (1).	1+1+1
	OR: Parenchyma (1); Collenchyma (1); Sclerenchyma (1).	1+1+1
18	Control (1); Location (1); Structure (1).	1+1+1
	OR: Squamous (1); Columnar (1); Cuboidal (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Definition (1); Apical (1); Lateral (1); Intercalary (1); Features (1).	1+1+1+1+1
	OR: Neuron (1); Dendrites (1); Axon (1); Impulse (1); Function (1).	1+1+1+1+1
20	Parenchyma (1); Collenchyma (1); Sclerenchyma (1); Features (1); Functions (1).	1+1+1+1+1
	OR: Hinge (1); Ball and socket (1); Pivot (1); Fixed (1); Examples (1).	1+1+1+1+1

Mistakes to avoid:

- Saying that xylem conducts food and phloem conducts water; xylem: water and minerals, phloem: food.
- Calling xylem vessels and sieve tubes both living; xylem conducting elements are dead, sieve tubes are living but without a nucleus and depend on companion cells.
- Mixing up tendons and ligaments (tendon: muscle to bone; ligament: bone to bone).
- Saying cardiac muscle is voluntary or unbranched; it is involuntary, striated and branched.
- Confusing apical meristem (length), lateral meristem (girth) and intercalary meristem (regrowth at base); also writing that meristem cells have large vacuoles (they have dense cytoplasm and small or no vacuoles).
- Listing blood as an epithelial tissue; it is a connective tissue with a fluid matrix (plasma).

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 10: **Tissues in Action** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** connective tissue	Fluid.
2	**(a)** fat	Fat.
3	**(c)** epidermis of leaves	Gas exchange.
4	**(c)** connective tissue	Hard.
5	**(a)** columnar epithelium	Absorbs.
6	**(c)** cork	Protective.
7	**(c)** cardiac	Involuntary.
8	**(a)** ear	Flexible.
9	**(d)** sclerenchyma	Strong fibres.
10	**(a)** striated	Voluntary.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Tiny pores in the epidermis of leaves (2).	2
12	Tissue connecting muscle to bone (2).	2
13	Striated (1); Smooth and cardiac (1).	1+1
14	Protection (1); Prevents water loss (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Storage (1); Photosynthesis (chlorenchyma) (1); Floating (aerenchyma) (1).	1+1+1
16	Dendrites (1); Cell body (1); Axon (1).	1+1+1
17	Protection (1); Prevents water loss (1); Insulation (1).	1+1+1
	OR: Protection (1); Absorption (1); Secretion (1).	1+1+1
18	Hinge: elbow (1); Ball and socket: hip (1); Pivot: neck (1).	1+1+1
	OR: Epidermis (1); Xylem (1); Phloem (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Xylem parts (1); Xylem function (1); Phloem parts (1); Phloem function (1); Difference (1).	1+1+1+1+1
	OR: Growth (1); Types (1); Support (1); Conduction (1); Examples (1).	1+1+1+1+1
20	Squamous (1); Cuboidal (1); Columnar (1); Ciliated (1); Functions (1).	1+1+1+1+1
	OR: Epidermis (1); Cuticle (1); Stomata (1); Root hair (1); Cork (1).	1+1+1+1+1

Mistakes to avoid:

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- Calling xylem vessels and sieve tubes both living; xylem conducting elements are dead, sieve tubes are living but without a nucleus and depend on companion cells.
- Mixing up tendons and ligaments (tendon: muscle to bone; ligament: bone to bone).
- Saying cardiac muscle is voluntary or unbranched; it is involuntary, striated and branched.
- Confusing apical meristem (length), lateral meristem (girth) and intercalary meristem (regrowth at base); also writing that meristem cells have large vacuoles (they have dense cytoplasm and small or no vacuoles).
- Listing blood as an epithelial tissue; it is a connective tissue with a fluid matrix (plasma).

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ANSWER KEY & MARKING SCHEME SET 3

Chapter 10: **Tissues in Action** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** sclerenchyma	Lignin.
2	**(b)** parenchyma	Storage.
3	**(d)** neuron	Nerve cell.
4	**(a)** protection	Covering.
5	**(a)** collenchyma	Flexible.
6	**(c)** bone to bone	Elastic.
7	**(d)** absorption	Water.
8	**(a)** length	Tips.
9	**(d)** muscle to bone	Strong.
10	**(a)** open and close stomata	Regulate.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Flexible connective tissue (2).	2
12	Conducting tissue carrying water (2).	2
13	Nerve cell (2).	2
14	Hinge: knee (1); Ball and socket: shoulder (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Blood (1); Bone (1); Cartilage (1).	1+1+1
16	Direction (1); Material carried (1); Living or dead cells (1).	1+1+1
17	Stores fat (1); Insulation (1); Cushions organs (1).	1+1+1
	OR: Support (1); Protection (1); Movement (1).	1+1+1
18	Strength (1); Support (1); Protection (1).	1+1+1
	OR: Cloning (1); Disease-free plants (1); Research (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Blood (1); Bone (1); Cartilage (1); Ligament (1); Tendon (1).	1+1+1+1+1
	OR: Onion roots (1); Water (1); Measure (1); Tip growth (1); Conclusion (1).	1+1+1+1+1
20	Striated (1); Smooth (1); Cardiac (1); Voluntary (1); Involuntary (1).	1+1+1+1+1
	OR: Medium (1); Single cell (1); Plant (1); Uses (1); Steward (1).	1+1+1+1+1

Mistakes to avoid:

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